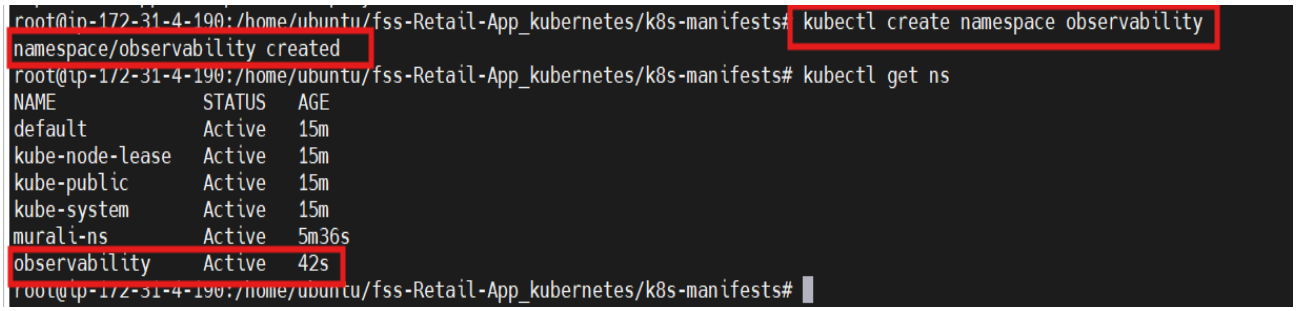
ELK

PREREQUISITES

```
kubectl get nodes
```

Create namespace:

```
kubectl create namespace observability
```



PART 1: LOGS → ELK STACK ON EKS

STEP 1 Elasticsearch

- Create a file called deployment-elasticsearch.yaml and paste below script inside the file 👇

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: elasticsearch-deployment
  namespace: observability
spec:
  replicas: 1
  selector:
    matchLabels:
      app: elasticsearch
  template:
    metadata:
      labels:
        app: elasticsearch
    spec:
      containers:
      - name: elasticsearch
        image: docker.elastic.co/elasticsearch/elasticsearch:8.11.0
        env:
```

```
        - name: discovery.type
          value: single-node
        - name: xpack.security.enabled
          value: "false"
        - name: ES_JAVA_OPTS
          value: "-Xms512m -Xmx512m"
        ports:
        - containerPort: 9200
```

```
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# vi deplotment-elasticsearch.yaml
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# ls
deplotment-elasticsearch.yaml  mongodb-deployment.yml  mongodb-service.yml  usernode-js-service.yml  userprofile-deployment.yml
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# kubectl apply -f deplotment-elasticsearch.yaml
deployment.apps/elasticsearch created
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# kubectl get all -n observability
NAME                                READY   STATUS              RESTARTS   AGE
pod/elasticsearch-cd7d4b6c5-pnwsg   0/1     ContainerCreating   0          15s

NAME                            READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/elasticsearch   0/1     1            0           15s

NAME                                       DESIRED   CURRENT   READY   AGE
replicaset.apps/elasticsearch-cd7d4b6c5    1         1         0       15s
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# cat deplotment-elasticsearch.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: elasticsearch
  namespace: observability
spec:
  replicas: 1
  selector:
    matchLabels:
      app: elasticsearch
  template:
    metadata:
      labels:
        app: elasticsearch
    spec:
      containers:
      - name: elasticsearch
        image: docker.elastic.co/elasticsearch/elasticsearch:8.11.0
        env:
        - name: discovery.type
          value: single-node
        - name: ES_JAVA_OPTS
          value: "-Xms512m -Xmx512m"
        ports:
        - containerPort: 9200

root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests#
```

- Create a file called svc.elasticsearch.yaml and paste below script inside the file 👇

Elasticsearch Service YAML (ClusterIP)

```
apiVersion: v1
kind: Service
metadata:
  name: elasticsearch
  namespace: observability
spec:
  type: ClusterIP
  selector:
    app: elasticsearch
  ports:
  - name: http
    port: 9200
    targetPort: 9200
```

```
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# vi svc-elasticsearch.yaml
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# ls
deplotment-elasticsearch.yaml  mongodb-deployment.yml  mongodb-service.yml  svc-elasticsearch.yaml  usernode-js-service.yml  userprofile-deployment.yml
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# cat svc-elasticsearch.yaml
apiVersion: v1
kind: Service
metadata:
  name: elasticsearch
  namespace: observability
spec:
  type: ClusterIP
  selector:
    app: elasticsearch
  ports:
  - name: http
    port: 9200
    targetPort: 9200

root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# kubectl apply -f svc-elasticsearch.yaml
service/elasticsearch created
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# kubectl get svc -n observability
NAME            TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)    AGE
elasticsearch   ClusterIP   10.100.20.31   <none>        9200/TCP   13s
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests#
```

Note: Elasticsearch should not be exposed publicly. It is accessed internally by Logstash and Kibana.

STEP 2️⃣ Logstash (Collect Pod Logs)

- Create a file called configmap-logstash.yaml and paste below script inside the file 👇

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: logstash-config
  namespace: observability
data:
  logstash.conf: |
    input {
      file {
        path => "/var/log/pods/**/0.log"
        start_position => "beginning"
      }
    }
    filter {
      json {
        source => "message"
      }
    }
    output {
      elasticsearch {
        hosts => ["http://elasticsearch:9200"]
        index => "k8s-logs"
      }
    }
```

```
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# vi configmap-elasticsearch.yaml
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# ls
configmap-elasticsearch.yaml  mongodb-deployment.yml  svc-elasticsearch.yaml   userprofile-deployment.yml
deplotment-elasticsearch.yaml  mongodb-service.yml     usernode-js-service.yml
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# cat configmap-elasticsearch.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: logstash-config
  namespace: observability
data:
  logstash.conf: |
    input {
      file {
        path => "/var/log/containers/*.log"
        start_position => "beginning"
      }
    }
    filter {
      json {
        source => "message"
      }
    }
    output {
      elasticsearch {
        hosts => ["http://elasticsearch:9200"]
        index => "k8s-logs"
      }
    }

root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# kubectl apply -f configmap-elasticsearch.yaml
configmap/logstash-config created
```

```
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# kubectl get cm -n observability
NAME               DATA   AGE
kube-root-ca.crt   1      24m
logstash-config    1      35s
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests#
```

- Create a file called deployment-logstash.yaml and paste below script inside the file 👇

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: logstash-deployment
  namespace: observability
spec:
  replicas: 1
  selector:
    matchLabels:
      app: logstash
  template:
    metadata:
      labels:
        app: logstash
    spec:
      containers:
      - name: logstash
        image: docker.elastic.co/logstash/logstash:8.11.0
        volumeMounts:
        - name: varlog
          mountPath: /var/log/pods
        - name: config
          mountPath: /usr/share/logstash/pipeline
      volumes:
      - name: varlog
        hostPath:
          path: /var/log/pods
      - name: config
```

```
        configMap:
          name: logstash-config
```

```
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# ls
configmap-elasticsearch.yaml  deployment-logstash.yaml  mongodb-service.yml     usernode-js-service.yml
deplotment-elasticsearch.yaml  mongodb-deployment.yml    svc-elasticsearch.yaml  userprofile-deployment.yml
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# cat deployment-logstash.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: logstash-deployment
  namespace: observability
spec:
  replicas: 1
  selector:
    matchLabels:
      app: logstash
  template:
    metadata:
      labels:
        app: logstash
    spec:
      containers:
      - name: logstash
        image: docker.elastic.co/logstash/logstash:8.11.0
        volumeMounts:
        - name: varlog
          mountPath: /var/log/containers
        - name: config
          mountPath: /usr/share/logstash/pipeline
      volumes:
      - name: varlog
        hostPath:
          path: /var/log/containers
      - name: config
        configMap:
          name: logstash-config

root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# kubectl apply -f deployment-logstash.yaml
deployment.apps/logstash-deployment created
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# kubectl get deployment -n observability
NAME                   READY   UP-TO-DATE   AVAILABLE   AGE
elasticsearch          1/1     1            1           22m
logstash-deployment    1/1     1            1           62s
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests#
```

A Service is NOT required for this Logstash deployment.

Why Service is NOT required for Logstash (in your case)

Let's look at **what your Logstash is doing** 👇

```
input {
  file {
    path => "/var/log/pods/**/0.log"
  }
}
```

✓ Logstash behavior here

- Reads logs **directly from node filesystem**
- Uses `hostPath: /var/log/containers`
- **Does NOT receive network traffic**
- **Does NOT expose any port**

👉 Because of this:

❌ No inbound traffic
❌ No clients connecting to Logstash
❌ No port listening

Kubernetes Service is unnecessary

STEP 3 Kibana

- Create a file called deployment-kibana.yaml and paste below script inside the file 👇

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: kibana-deployment
  namespace: observability
spec:
  replicas: 1
  selector:
    matchLabels:
      app: kibana
  template:
    metadata:
      labels:
        app: kibana
    spec:
      containers:
      - name: kibana
        image: docker.elastic.co/kibana/kibana:8.11.0
        env:
        - name: ELASTICSEARCH_HOSTS
          value: "http://elasticsearch:9200"
        ports:
        - containerPort: 5601
```

```
root@ip-172-31-4-190:/home/ubuntu# cd fss-Retail-App_kubernetes/k8s-manifests/
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# ls
configmap-logstash.yaml        deployment-logstash.yaml  mongodb-service.yml      usernode-js-service.yml
deployment-elasticsearch.yaml  mongodb-deployment.yml    svc-elasticsearch.yaml  userprofile-deployment.yml
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# vi deployment-kibana.yaml
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# ls
configmap-logstash.yaml        deployment-kibana.yaml    mongodb-deployment.yml  svc-elasticsearch.yaml   userprofile-deployment.yml
deployment-elasticsearch.yaml  deployment-logstash.yaml  mongodb-service.yml      usernode-js-service.yml
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# cat deployment-kibana.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: kibana-deployment
  namespace: observability
spec:
  replicas: 1
  selector:
    matchLabels:
      app: kibana
  template:
    metadata:
      labels:
        app: kibana
    spec:
      containers:
      - name: kibana
        image: docker.elastic.co/kibana/kibana:8.11.0
        env:
        - name: ELASTICSEARCH_HOSTS
          value: "http://elasticsearch:9200"
        ports:
        - containerPort: 5601

root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# kubectl apply -f deployment-kibana.yaml
deployment.apps/kibana-deployment created
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# kubectl get deployment -n observability
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
elasticsearch         1/1     1            1           35m
kibana-deployment     0/1     1            0           12s
logstash-deployment   1/1     1            1           14m
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests#
```

Kibana Service YAML (LoadBalancer)

- Create a file called svc-kibana.yaml and paste below script inside the file 👇

```
apiVersion: v1
kind: Service
metadata:
  name: kibana
  namespace: observability
spec:
  type: LoadBalancer
  selector:
    app: kibana
  ports:
  - name: http
    port: 5601
    targetPort: 5601
```

```
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# vi svc-kibana.yaml
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# ls
configmap-logstash.yaml        deployment-kibana.yaml    mongodb-deployment.yml  svc-elasticsearch.yaml  usernode-js-service.yml
deployment-elasticsearch.yaml  deployment-logstash.yaml  mongodb-service.yml     svc-kibana.yaml         userprofile-deployment.yml
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# cat svc-kibana.yaml
apiVersion: v1
kind: Service
metadata:
  name: kibana
  namespace: observability
spec:
  type: LoadBalancer
  selector:
    app: kibana
  ports:
  - name: http
    port: 5601
    targetPort: 5601

root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# kubectl apply -f svc-kibana.yaml
service/kibana created
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# kubectl get svc -n obersvability
No resources found in obersvability namespace.
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests# kubectl get svc -n observability
NAME            TYPE           CLUSTER-IP       EXTERNAL-IP                                                              PORT(S)          AGE
elasticsearch   ClusterIP      10.100.20.31     <none>                                                                   9200/TCP         36m
kibana          LoadBalancer   10.100.207.117   a8b8fc49b16ef418c9324f15a7393abe-74060292.sa-east-1.elb.amazonaws.com   5601:31109/TCP   25s
root@ip-172-31-4-190:/home/ubuntu/fss-Retail-App_kubernetes/k8s-manifests#
```

STEP 3 – Install Filebeat (Recommended)

Create file:

📄 **filebeat.yaml**

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: filebeat-config
  namespace: observability
data:
  filebeat.yml: |
    filebeat.autodiscover:
      providers:
        - type: kubernetes
          node: ${NODE_NAME}
          hints.enabled: true

    processors:
```

```
      - add_cloud_metadata: {}
      - add_host_metadata: {}
      - add_kubernetes_metadata: {}

    output.elasticsearch:
      hosts: ["http://elasticsearch-svc:9200"]

---
apiVersion: apps/v1
kind: DaemonSet
metadata:
  name: filebeat
  namespace: observability
spec:
  selector:
    matchLabels:
      app: filebeat
  template:
    metadata:
      labels:
        app: filebeat
    spec:
      serviceAccountName: default
      containers:
        - name: filebeat
          image: docker.elastic.co/beats/filebeat:8.11.0
          args: ["-c", "/etc/filebeat.yml", "-e"]
          env:
            - name: NODE_NAME
              valueFrom:
                fieldRef:
                  fieldPath: spec.nodeName
          volumeMounts:
            - name: config
              mountPath: /etc/filebeat.yml
              subPath: filebeat.yml
            - name: varlog
              mountPath: /var/log
      volumes:
        - name: config
          configMap:
            name: filebeat-config
        - name: varlog
          hostPath:
            path: /var/log
```

Apply:

```
kubectl apply -f filebeat.yaml
```

How to create connecter in Kibana

Configure Encryption Key in Kibana

You must update your **Kibana deployment config**.

Since you're running Kibana in Kubernetes, this means:

👉 Add environment variable in Kibana Deployment.

STEP 1 — Generate Encryption Key

On your terminal:

```
openssl rand -base64 32
```

Example output:

```
Gk3kD1A7sdf23...==
```

Copy this value.

WHY THIS IS REQUIRED

Kibana stores:

- Slack webhook URLs
- Email passwords
- API tokens

These must be encrypted on disk.

So Kibana refuses to enable alerting until you define an encryption key.

Add These Environment Variables

Inside the container env section, add:

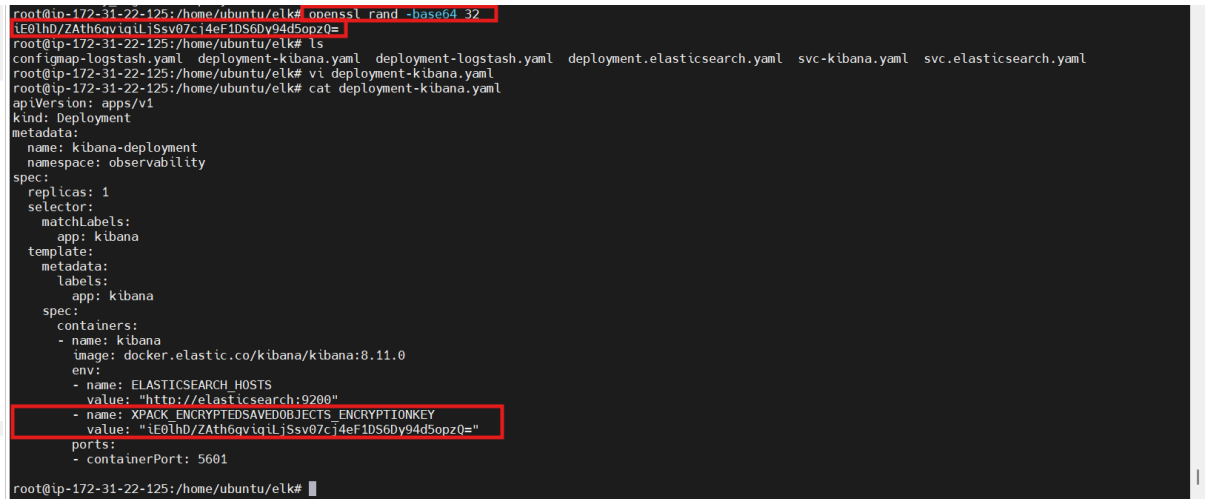
```
env:
  - name: XPACK_ENCRYPTEDSAVEDOBJECTS_ENCRYPTIONKEY
    value: "PASTE_YOUR_GENERATED_KEY_HERE"
```

Example:

```
env:
  - name: XPACK_ENCRYPTEDSAVEDOBJECTS_ENCRYPTIONKEY
    value: "Gk3kD1A7sdf23...=="
```

⚠ Must be minimum 32 characters.

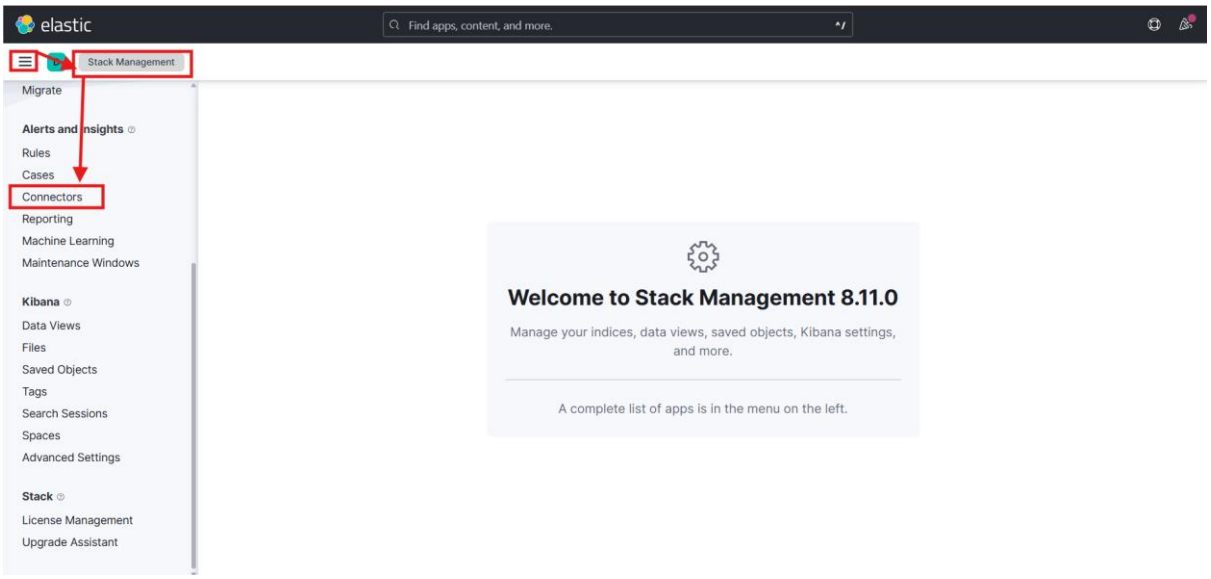
Save and exit.

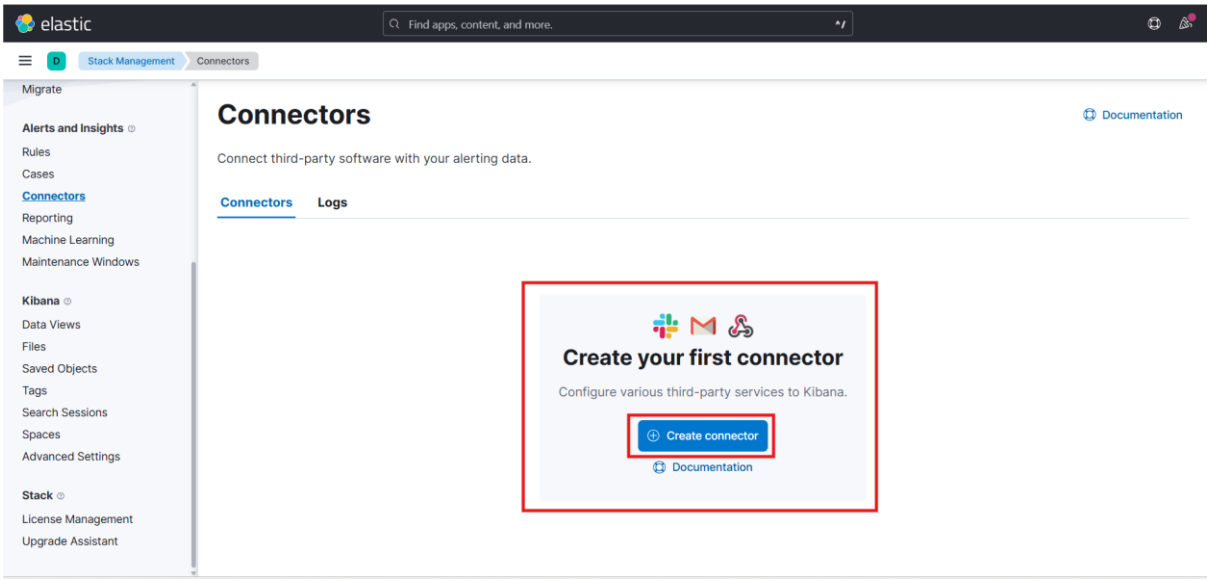


Add Slack Connector in Kibana

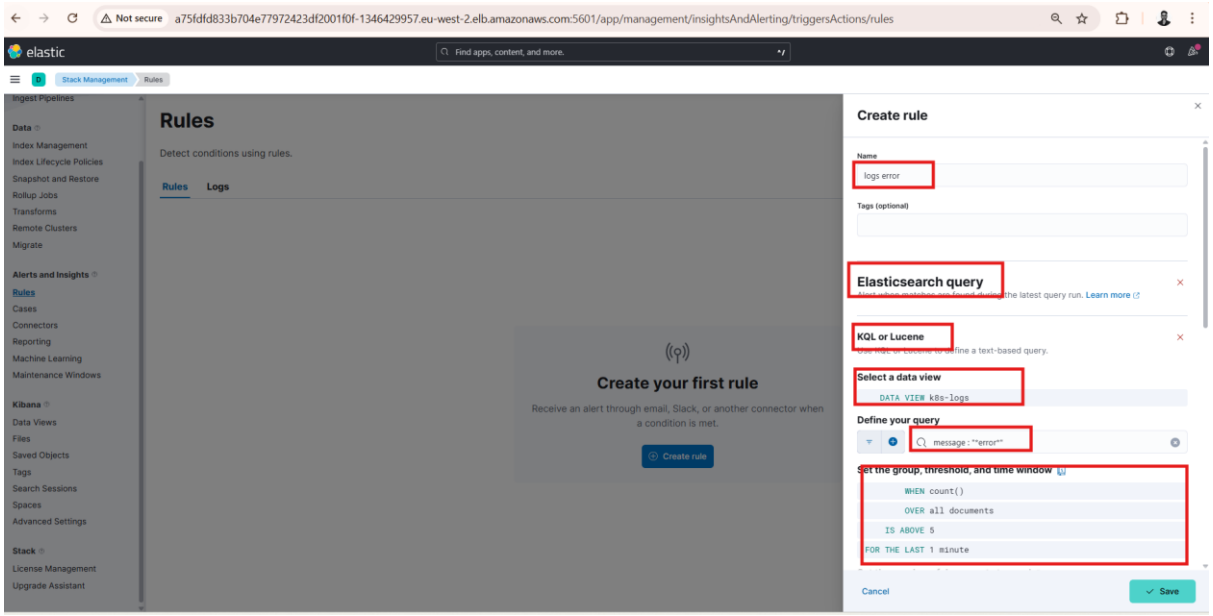
In Kibana:

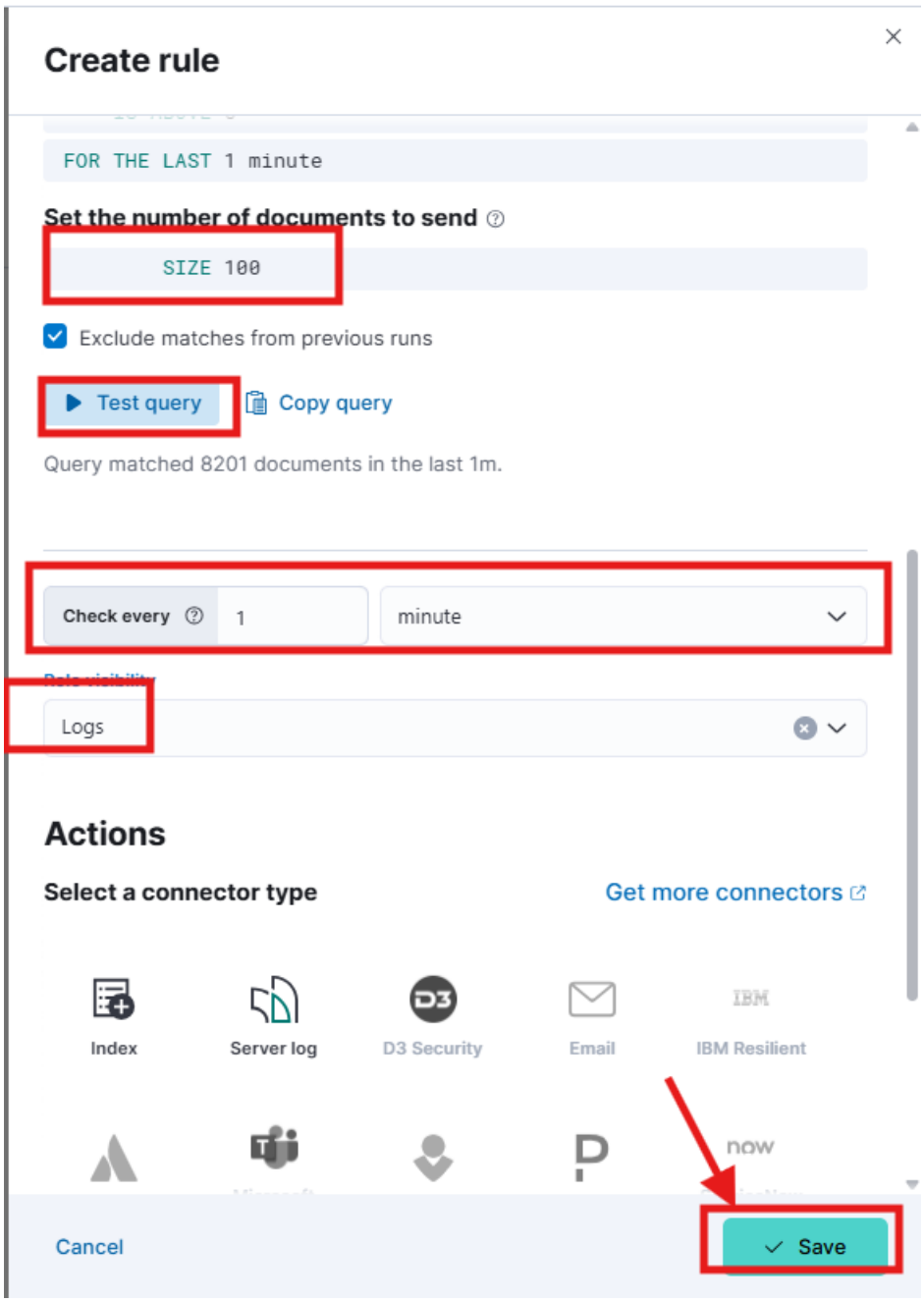
→ Stack Management
→ Connectors
→ Create connector

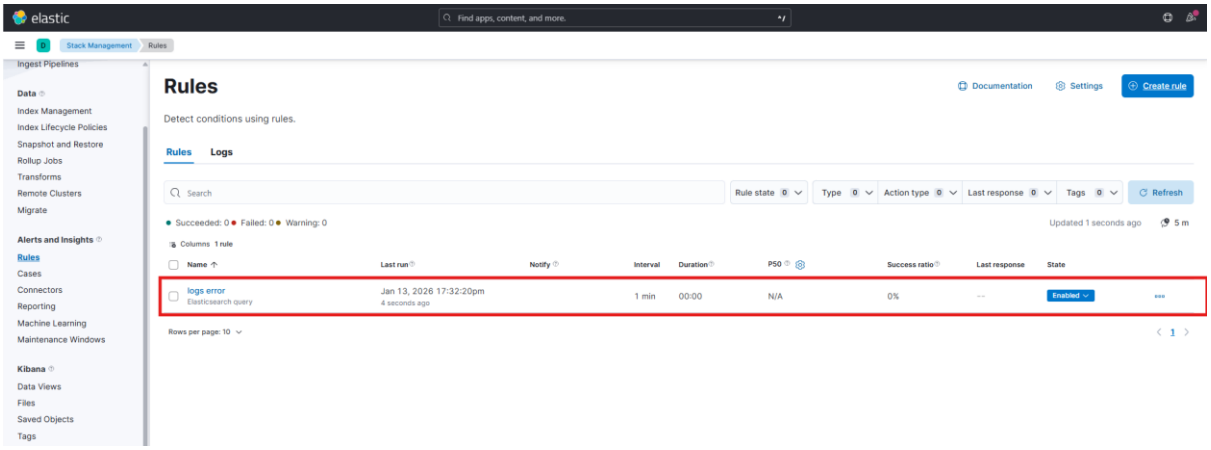




How to create alert rules




Create rule
FOR THE LAST 1 minute
Set the number of documents to send
SIZE 100
Exclude matches from previous runs
Test query
Copy query
Query matched 8201 documents in the last 1m.
Check every
1
minute
Logs
Actions
Select a connector type
Get more connectors
Index
Server log
D3 Security
Email
IBM Resilient
now
Cancel
Save


elastic
Find apps, content, and more.
Stack Management
Rules
Ingest Pipelines
Data
Index Management
Index Lifecycle Policies
Snapshot and Restore
Rollup Jobs
Transforms
Remote Clusters
Migrate
Alerts and Insights
Rules
Cases
Connectors
Reporting
Machine Learning
Maintenance Windows
Kibana
Data Views
Files
Saved Objects
Tags
Rules
Documentation
Settings
Create rule
Detect conditions using rules.
Rules
Logs
Search
Rule state 0
Type 0
Action type 0
Last response 0
Tags 0
Refresh
Succeeded: 0 Failed: 0 Warning: 0
Updated 1 seconds ago
5 m
Columns 1 rule
Name
Last run
Notify
Interval
Duration
PSO
Success ratio
Last response
State
logs error
Elasticsearch query
Jan 13, 2026 17:32:20pm
4 seconds ago
1 min
00:00
N/A
0%
--
Enabled
Rows per page: 10
1